

Model Progress Report: Decision Tree Classifier

Task ID: TB-L10

Date: 05-Sep

Algorithm: CART (Gini Impurity)

COMPLETED

HOLDOUT TEST ACCURACY

80.00%

Evaluated on 200 test samples

OPTIMAL ARCHITECTURE

max_depth = 3

Tuned across range 1 to 15

DOMINANT PREDICTOR

Credit Score

30.42% relative importance

1. MODEL OBJECTIVE & ARCHITECTURE

- Algorithm Implementation:** DecisionTreeClassifier based on the CART framework optimizing split points with Gini impurity.
- Feature Pipeline:** Directly consumes **13 one-hot encoded features** without requiring feature standardization or scaling.
- Dataset Partition:** Stratified 80/20 train-test partition (**800 training rows, 200 testing rows**) with fixed random_state=42.

2. HYPERPARAMETER TUNING & OVERFITTING CONTROL

- Depth Exploration:** Explored tree depths from max_depth = 1 to 15 to map out the bias-variance tradeoff curve.
- Optimal Depth Selection:** Constrained tree growth to max_depth = 3 to preserve high generalizability, prevent overfitting, and enable direct regulatory interpretability.
- Generalization Performance:** Achieved a solid baseline test accuracy of **80.00%** on the unseen test set.

3. FEATURE IMPORTANCE PROFILE

Feature Name	Importance (%)	Relative Influence
Credit_Score	30.42%	
Loan_Amount_Requested	17.65%	
Annual_Income	16.29%	
Monthly_Expenses	9.99%	
Loan_Term	9.42%	
Outstanding_Debt	7.17%	

4. OUTPUT ARTIFACTS PRODUCED



decision_tree_model.pkl

Trained DecisionTreeClassifier instance (max_depth=3)



model_columns.pkl

Column alignment schema (13 features)



Diagnostic Plots

Tree schematic visualization (plot_tree) and Feature Importance bar chart